

Selected Topics Assignment #1

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Q1)

x	2	6	8	10	16	20
y	1.41	2.24	3.16	5	9	11.2

a) $y = ax + b$

$$1.41 = 2a + b$$

$$2.24 = 6a + b$$

$$a = 0.2075 \quad b = 0.995$$

$$y = 0.2075x + 0.995 \rightarrow x = 4 \rightarrow y = 1.825$$

b) $y = ax + b$

$$9 = 16a + b$$

$$11.2 = 20a + b \quad a = 0.55 \quad b = 0.2$$

$$y = 0.55x + 0.2 \rightarrow x = 24 \rightarrow y = 13.4$$

c) $y = ax^2 + bx + c$

$$5 = 100a + 10b + c$$

$$9 = 256a + 16b + c$$

$$11.2 = 400a + 20b + c$$

$$a = -7/600 \quad b = 0.97 \quad c = -53/15$$

$$y = -7/600x^2 + 0.97x - 53/15 \rightarrow @ x = 24 \rightarrow y = 977/75$$
$$y = 13.027$$

d) $5 = 100a + 10b + c$

$$9 = 256a + 16b + c$$

$$11.2 = 400a + 20b + c$$

$$a = -7/600 \quad b = 0.97 \quad c = -53/15$$

$$y = -7/600x^2 + 0.97x - 53/15 \rightarrow @ y = 15.8 \rightarrow x = 33.14$$

or $x = 50$

Q2)

Linear extrapolation

use points $(2, 5.76)$ and $(8.25, 9.71)$

$$y = ax + b$$

$$5.76 = 2a + b$$

$$9.71 = 8.25a + b$$

$$a = 0.632 \quad b = 4.496$$

$$y = 0.632x + 4.496$$

$$@ y = 11 \quad 11 = 0.632x + 4.496 \quad x = 10.29$$

polynomial extrapolation

$(1.5, 3.41)$ & $(2, 5.76)$ & $(8.25, 9.71)$

$$y = ax^2 + bx + c$$

$$3.41 = 2.25a + 1.5b + c$$

$$5.76 = 4a + 2b + c$$

$$9.71 = 68.0625a + 8.25b + c$$

$$a = -0.6 \quad b = 6.81 \quad c = -5.448$$

$$y = -0.6x^2 + 6.81x - 5.448$$

$$@ y = 11 \quad 11 = -0.6x^2 + 6.81x - 5.448$$

$$x = 7.864 \text{ or } x = 3.49$$